

INTCK

Say exactly what you are counting

01 / THE SAS EXAMPLE

```
data test;
  input start_date :date9. end_date :date9.;
  days = intck('day', start_date, end_date);
  months = intck('month', start_date, end_date);
  years = intck('year', start_date, end_date);
  format start_date end_date date9.;
  datalines;
01JAN2023 15JAN2023
01JAN2023 01FEB2023
01JAN2023 31DEC2023
15MAR2023 20MAR2023
31JAN2023 01FEB2023
31DEC2023 01JAN2024
;
run;
proc print data=test noobs;
run;
```

ATTACH SOURCE OR PASTE IT AFTER THE PROMPT. START FRESH EACH TIME.

WHAT YOU WILL PRACTICE

Turn a familiar business rule into an explicit conversion requirement.

Default INTCK counts calendar boundaries. Think of turning a calendar page: January 31 to February 1 crosses a month boundary, although only one day passes. Compare all six rows.

EXPECTED RESULT

start_date	end_date	days	months	years
01JAN2023	15JAN2023	14	0	0
01JAN2023	01FEB2023	31	1	0
01JAN2023	31DEC2023	364	11	0
15MAR2023	20MAR2023	5	0	0
31JAN2023	01FEB2023	1	1	0
31DEC2023	01JAN2024	1	1	1

The last two rows distinguish calendar boundaries from elapsed months or years. Compare every result column.

Expected values checked in Python; SAS execution pending.

Prompt A: leave choices open

High degrees of freedom / Notice the decisions the model makes for you.

COPY THIS PROMPT + THE SAS EXAMPLE

Convert this SAS code to Python.
Show code in this response. Do not create or run files, or read existing Python files.

LOOK AT THE RESPONSE

Which library and method did it choose? Did it preserve the expected behavior? Compare the result with page 1 before accepting it.

TRY IT / THREE FRESH ATTEMPTS

Run A three times with the same SAS example and setup. Save each first response before asking follow-up questions. Record agreement as well as differences.

Bedrock A/B trials for this example are pending.

WHY CHOICES REMAIN OPEN

- 1 A clear destination**
Naming Python sets the target language. The source and input rows give the model a concrete task.
- 2 An unstated meaning**
The model must recognize boundary counting. A plausible duration calculation could answer a different question.
- 3 Open implementation choices**
Library and method remain open. Different code can still be correct; compare the actual results.

ANTI-PATTERN TO AVOID

Anti-pattern: "Calculate months between dates." Completed months and crossed month boundaries are different rules. State the meaning and include a one-day boundary example.

Prompt B: make the choices explicit

Low degrees of freedom / State the behavior, then ask for a check.

COPY THIS PROMPT + THE SAME SAS EXAMPLE

```
Convert this SAS code to Python using pandas.  
Show code in this response. Do not create or run files, or read  
existing Python files.
```

- Recreate the six rows in order; keep the same column names.
- Parse dates with `pd.to_datetime(..., format='%d%b%Y')`.
The inputs are valid dates with English month names.
- Preserve default discrete INTCK behavior: count boundaries.
- `days`: subtract `start_date` from `end_date` and use `.dt.days`.
- `months`: `12 * (end year - start year) + end month - start month`.
- `years`: `end year - start year`.
- Return one complete script and include assertions for:
`days = [14, 31, 364, 5, 1, 1]`
`months = [0, 1, 11, 0, 1, 1]`
`years = [0, 0, 0, 0, 0, 1]`
- Calculate from the input dates; do not hard-code result columns.
Briefly explain why a one-day interval can cross a month or year.

WHY THESE DETAILS HELP

- 1

Define the meaning
Boundary counting specifies the question. Turning a calendar page makes the rule easy to explain.
- 2

Choose the route
Naming pandas and the formulas narrows choices without requiring you to write the whole program.
- 3

Make success visible
Complete expected lists check every row. One-day boundary examples expose an incorrect duration calculation.

CHECK IT YOURSELF

Run the script separately. Compare all three columns. January 31 to February 1 must cross one month; December 31 to January 1 must cross one year.

Name the rule behind a calculation and include a small example that distinguishes it from another plausible rule.